

Q.1) Write HTML script to display the text in bold, italic, underline and with strike. Apply separate effect on different text.

```
<!DOCTYPE html>
<html>
<head>
  <title>Text Formatting</title>
</head>
<body>

  <h2>Text Formatting Example</h2>

  <p><b>This text is in Bold</b></p>

  <p><i>This text is in Italic</i></p>

  <p><u>This text is Underlined</u></p>

  <p><strike>This text has Strike-through</strike></p>

</body>
</html>
```

Q.2) Write a menu driven program to perform the following operations on associative arrays:

- a) Merge the given arrays.**
- b) Find the intersection of two arrays.**
- c) Find the union of two arrays.**
- d) Find set difference of two arrays.**

```
<?php

function mergeArrays($arr1, $arr2) {
  return array_merge($arr1, $arr2);
}

function intersectArrays($arr1, $arr2) {
  return array_intersect_assoc($arr1, $arr2);
}

function unionArrays($arr1, $arr2) {
```

```

    return $arr1 + $arr2; // union
}

function differenceArrays($arr1, $arr2) {
    return array_diff_assoc($arr1, $arr2);
}

// Sample associative arrays
$array1 = array("a"=>1, "b"=>2, "c"=>3);
$array2 = array("b"=>2, "c"=>4, "d"=>5);

echo "Menu:\n";
echo "1. Merge Arrays\n";
echo "2. Intersection\n";
echo "3. Union\n";
echo "4. Difference\n";

echo "Enter your choice: ";
$choice = trim(fgets(STDIN));

switch($choice) {
    case 1:
        print_r(mergeArrays($array1, $array2));
        break;

    case 2:
        print_r(intersectArrays($array1, $array2));
        break;

    case 3:
        print_r(unionArrays($array1, $array2));
        break;

    case 4:
        print_r(differenceArrays($array1, $array2));
        break;

    default:
        echo "Invalid choice!";
}

```

?>

Q.1) Create an html page with following specifications :

a. Title should be about “My City”

b. Place your City name at the top of the page in large text and in red color

c. Add names and images (as a link) of landmarks in your city each in a different color, style and typeface .

d. After clicking on images it should display history of that place.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>My City</title>
```

```
</head>
```

```
<body>
```

```
    <!-- City Name -->
```

```
    <h1 style="color:red; font-size:50px; text-align:center;">
```

```
        Nashik
```

```
    </h1>
```

```
    <!-- Landmark 1 -->
```

```
    <h2 style="color:blue; font-family:Arial;">
```

```
        Panchavati
```

```
    </h2>
```

```
    <a href="panchavati.html">
```

```
        
```

```
    </a>
```

```
    <!-- Landmark 2 -->
```

```
    <h2 style="color:green; font-family:Courier;">
```

```
        Sula Vineyards
```

```
    </h2>
```

```
    <a href="sula.html">
```

```
        
```

```
    </a>
```

```
    <!-- Landmark 3 -->
```

```
    <h2 style="color:purple; font-family:Georgia;">
```

```
        Trimbakeshwar Temple
```

```
    </h2>
```

```
    <a href="trimbak.html">
```

```

</a>
```

```
</body>
</html>
```

QQ1 Write a PHP program to create a class temperature which contains data members as Celsius and Fahrenheit . Create and Initialize all values of temperature object by using parameterized constructor . Convert Celsius to Fahrenheit and Convert Fahrenheit to Celsius using member functions. Display conversion on next page.

```
<?php
```

```
class Temperature {
    public $celsius;
    public $fahrenheit;

    // Parameterized constructor
    function __construct($c, $f) {
        $this->celsius = $c;
        $this->fahrenheit = $f;
    }

    // Convert Celsius to Fahrenheit
    function cToF() {
        return ($this->celsius * 9/5) + 32;
    }

    // Convert Fahrenheit to Celsius
    function fToC() {
        return ($this->fahrenheit - 32) * 5/9;
    }
}

// Get values from form
$c = $_POST['celsius'];
$f = $_POST['fahrenheit'];

// Create object using parameterized constructor
$temp = new Temperature($c, $f);
```

?>

<!DOCTYPE html>

<html>

<head>

<title>Conversion Result</title>

</head>

<body>

<h2>Temperature Conversion Result</h2>

<?php

echo "Celsius to Fahrenheit: " . \$temp->cToF() . " °F
";

echo "Fahrenheit to Celsius: " . \$temp->fToC() . " °C";

?>

</body>

</html>

Q.1) Write HTML code to display following output.

(15)

▪ **Tea**

o **Hot tea**

o **Black tea**

▪ **Coffee**

• **Cold coffee**

• **Hot coffee**

<!DOCTYPE html>

<html>

<head>

<title>Tea and Coffee List</title>

</head>

<body>

<h2>Beverages</h2>

<ul type="square">

Tea

<ul type="circle">

```

        <li>Hot tea</li>
        <li>Black tea</li>
    </ul>
</li>

    <li>Coffee
        <ul type="disc">
            <li>Cold coffee</li>
            <li>Hot coffee</li>
        </ul>
    </li>
</ul>

```

```

</body>
</html>

```

Declare array. Reverse the order of elements, making the first element last and last element first and similarly rearranging other array elements.

[Hint : array_reverse()]

```
<?php
```

```

// Declare an array
$arr = array(10, 20, 30, 40, 50);

```

```

// Display original array
echo "Original Array: <br>";
print_r($arr);

```

```

// Reverse the array
$reversed = array_reverse($arr);

```

```

// Display reversed array
echo "<br><br>Reversed Array: <br>";
print_r($reversed);

```

```
?>
```

Q.1) Write HTML code to display the list of different courses available in our college using ordered as well as unordered list.

```

<!DOCTYPE html>
<html>

```

```

<head>
  <title>Courses in Our College</title>
</head>
<body>

<h2>Courses Available in Our College</h2>

<ol>
  <li>Undergraduate Courses
    <ul>
      <li>B.Sc (Computer Science)</li>
      <li>B.Com</li>
      <li>B.A</li>
    </ul>
  </li>

  <li>Postgraduate Courses
    <ul>
      <li>M.Sc (Computer Science)</li>
      <li>M.Com</li>
      <li>M.A</li>
    </ul>
  </li>

  <li>Diploma Courses
    <ul>
      <li>Diploma in Web Development</li>
      <li>Diploma in Banking</li>
    </ul>
  </li>
</ol>

</body>
</html>

```

Q.2) Write PHP script to demonstrate the concept of introspection for examining object.

```

<?php

class Student {

```

```

public $name = "Rahul";
public $age = 20;

function display() {
    echo "Student Name: " . $this->name;
}

function getAge() {
    return $this->age;
}
}

// Create object
$obj = new Student();

// Introspection functions
echo "<h3>Class Name:</h3>";
echo get_class($obj);

echo "<h3>Class Methods:</h3>";
print_r(get_class_methods($obj));

echo "<h3>Class Properties:</h3>";
print_r(get_class_vars(get_class($obj)));

echo "<h3>Object Variables:</h3>";
print_r(get_object_vars($obj));

?>

```

Q.1) Design a admission form. which should contains : text box, multiline text box, a table which shows your academic record, radio button, check box, submit button.

```

<!DOCTYPE html>
<html>
<head>
    <title>Admission Form</title>
</head>
<body>

```


<h2>College Admission Form</h2>

<form>

<!-- Text Box -->

Name: <input type="text" name="name">

<!-- Radio Buttons -->

Gender:

<input type="radio" name="gender" value="Male"> Male

<input type="radio" name="gender" value="Female"> Female

<!-- Checkboxes -->

Hobbies:

<input type="checkbox" name="hobby[]" value="Reading"> Reading

<input type="checkbox" name="hobby[]" value="Sports"> Sports

<input type="checkbox" name="hobby[]" value="Music"> Music

<!-- Multiline Text Box -->

Address:

<textarea name="address" rows="4" cols="40"></textarea>

<!-- Academic Record Table -->

<h3>Academic Record</h3>

<table border="1">

<tr>

<th>Exam</th>

<th>Board/University</th>

<th>Percentage</th>

</tr>

<tr>

<td>10th</td>

<td><input type="text" name="board10"></td>

<td><input type="text" name="per10"></td>

</tr>

<tr>

```

        <td>12th</td>
        <td><input type="text" name="board12"></td>
        <td><input type="text" name="per12"></td>
    </tr>
    <tr>
        <td>Graduation</td>
        <td><input type="text" name="boardgrad"></td>
        <td><input type="text" name="pergrad"></td>
    </tr>
</table>

```

```
<br>
```

```

<!-- Submit Button -->
<input type="submit" value="Submit">

```

```
</form>
```

```
</body>
```

```
</html>
```

Q.2) Write a PHP Script to accept customer Name from user and do the following

a) Transform Customer Name all Upper case latter.

Make First character to Upper Case.

```

<!DOCTYPE html>
<html>
<head>
    <title>Customer Name Transformation</title>
</head>
<body>

```

```

<form method="post">
    Enter Customer Name:
    <input type="text" name="custname">
    <input type="submit" value="Submit">
</form>

```

```

<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {

```

```

$name = $_POST['custname'];

echo "<h3>Results:</h3>";

// a) Convert to Uppercase
echo "Uppercase: " . strtoupper($name) . "<br>";

// b) First character uppercase
echo "First Letter Capital: " . ucfirst($name);
}
?>

</body>
</html>

```

Q.1) Write inline CSS program to display with background color pink with red colored text.

```

<!DOCTYPE html>
<html>
<head>
  <title>Inline CSS Example</title>
</head>
<body style="background-color: pink;">

  <h2 style="color: red;">This is a heading with red text</h2>

  <p style="color: red;">
    This page has a pink background and red colored text using inline CSS.
  </p>

</body>
</html>

```

Q.2) Write a menu driven program the following operation on an associative array

- a) Reverse the order of each element's key-value pair. [Hint: array_flip()]**
- b) Traverse the element in an array in random order. [Hint: shuffle()]**

```

<?php

```

```
// Declare associative array
$arr = array("a"=>10, "b"=>20, "c"=>30, "d"=>40);

echo "Menu:\n";
echo "1. Reverse key-value pairs\n";
echo "2. Traverse array in random order\n";

echo "Enter your choice: ";
$choice = trim(fgets(STDIN));

switch($choice) {

    case 1:
        echo "Original Array:\n";
        print_r($arr);

        // Reverse key-value pairs
        $flipped = array_flip($arr);

        echo "After Reversing Key-Value Pairs:\n";
        print_r($flipped);
        break;

    case 2:
        echo "Original Array:\n";
        print_r($arr);

        // Shuffle works on indexed arrays, so convert
        $temp = array_values($arr);
        shuffle($temp);

        echo "Random Order Traversal:\n";
        print_r($temp);
        break;

    default:
        echo "Invalid choice!";
}
?>
```

Q.1) Design a table which shows weekly time table of a specific class.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Weekly Time Table</title>
```

```
</head>
```

```
<body>
```

```
<h2 style="text-align:center;">FYBSc Weekly Time Table</h2>
```

```
<table border="1" cellspacing="0" cellpadding="10" align="center">
```

```
  <tr>
```

```
    <th>Day/Time</th>
```

```
    <th>10:00 - 11:00</th>
```

```
    <th>11:00 - 12:00</th>
```

```
    <th>12:00 - 01:00</th>
```

```
    <th>02:00 - 03:00</th>
```

```
  </tr>
```

```
  <tr>
```

```
    <th>Monday</th>
```

```
    <td>Math</td>
```

```
    <td>Physics</td>
```

```
    <td>Chemistry</td>
```

```
    <td>Lab</td>
```

```
  </tr>
```

```
  <tr>
```

```
    <th>Tuesday</th>
```

```
    <td>English</td>
```

```
    <td>Math</td>
```

```
    <td>Physics</td>
```

```
    <td>Lab</td>
```

```
  </tr>
```

```
  <tr>
```

```
    <th>Wednesday</th>
```

```
    <td>Chemistry</td>
```

```
    <td>English</td>
```

```

        <td>Math</td>
        <td>Sports</td>
    </tr>

    <tr>
        <th>Thursday</th>
        <td>Physics</td>
        <td>Chemistry</td>
        <td>English</td>
        <td>Lab</td>
    </tr>

    <tr>
        <th>Friday</th>
        <td>Math</td>
        <td>Physics</td>
        <td>Computer</td>
        <td>Library</td>
    </tr>

</table>

</body>
</html>

```

Q.2) Write a script to create XML file as 'Employee.xml'. The element of this xml file are as follows:

```

<Empdetails>
<Employee EMPno= Empname=>
<Salary>-----</Salary>
<Designation>-----</Designation>
</Employee>
</Empdetails>
<?php

```

```

// Create XML structure
$xml = new SimpleXMLElement("<Empdetails></Empdetails>");

// Add employee node with attributes
$employee = $xml->addChild("Employee");

```

```
$employee->addAttribute("EMPno", "101");
$employee->addAttribute("Empname", "Rahul");

// Add child elements
$employee->addChild("Salary", "50000");
$employee->addChild("Designation", "Developer");

// Save XML file
$xml->asXML("Employee.xml");

echo "Employee.xml file created successfully!";

?>
```

Q.1) Write internal CSS program to display with background color black with white colored text.

```
<!DOCTYPE html>
<html>
<head>
  <title>Internal CSS Example</title>

  <style>
    body {
      background-color: black;
      color: white;
    }

    h1 {
      color: white;
      text-align: center;
    }

    p {
      color: white;
      font-size: 18px;
    }
  </style>

</head>

<body>
```

```
<h1>Welcome to My Page</h1>
```

```
<p>
```

This page uses internal CSS with black background and white text color.

```
</p>
```

```
</body>
```

```
</html>
```

Q.2) Write a PHP script to accept Employee details (eno, ename, address) on first page. On second page accept earning (Basic, Da, HRA). On third page print Employee information(eno, ename, Address, BASIC, DA, HRA, TOTAL)

```
<?php
```

```
session_start();
```

```
?>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Employee Details</title>
```

```
</head>
```

```
<body>
```

```
<form method="post" action="emp2.php">
```

Employee No: <input type="text" name="eno">

Employee Name: <input type="text" name="ename">

Address: <textarea name="address"></textarea>


```
<input type="submit" value="Next">
```

```
</form>
```

```
</body>
```

```
</html>
```

Q.1 Write external CSS program to display with background color sky blue with blue colored text.

```
<!DOCTYPE html>
```

```
<html>
```



```

<head>
  <title>External CSS Example</title>
  <link rel="stylesheet" type="text/css" href="style.css">
</head>

<body>

  <h1>Welcome to External CSS Page</h1>

  <p>This page uses external CSS with sky blue background and blue text.</p>

</body>
</html>
body {
  background-color: skyblue;
  color: blue;
}

h1 {
  text-align: center;
  color: blue;
}

p {
  font-size: 18px;
}

```

Q.2)Write a Calculator class that can accept two values, then add them, subtract them,multiply them together, or divide them on request.

```

<?php

class Calculator {
  public $a;
  public $b;

  // Constructor
  function __construct($x, $y) {
    $this->a = $x;
    $this->b = $y;
  }
}

```

```
// Addition
function add() {
    return $this->a + $this->b;
}

// Subtraction
function subtract() {
    return $this->a - $this->b;
}

// Multiplication
function multiply() {
    return $this->a * $this->b;
}

// Division
function divide() {
    if ($this->b != 0)
        return $this->a / $this->b;
    else
        return "Division by zero error!";
}
}

// Example usage
$num1 = 20;
$num2 = 10;

// Create object
$calc = new Calculator($num1, $num2);

echo "Addition: " . $calc->add() . "<br>";
echo "Subtraction: " . $calc->subtract() . "<br>";
echo "Multiplication: " . $calc->multiply() . "<br>";
echo "Division: " . $calc->divide();

?>
```

Q.1) Write CSS using HTML which uses of text decoration, border, padding and margin.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Properties Demo</title>

  <style>
    .box {
      border: 3px solid blue;    /* border */
      padding: 20px;           /* space inside border */
      margin: 30px;            /* space outside border */
      text-decoration: underline; /* text decoration */
      font-size: 20px;
      color: darkblue;
    }

    .box2 {
      border: 2px dashed red;
      padding: 15px;
      margin: 20px;
      text-decoration: line-through;
      color: red;
    }
  </style>

</head>

<body>

  <div class="box">
    This is a text with underline, border, padding and margin.
  </div>

  <div class="box2">
    This is another text with strike-through and different styling.
  </div>

</body>
</html>
```

Q.2)Write a PHP script for the following. Design a form to accept a string and check whether the given string is Palindrome or not.

```
<!DOCTYPE html>
<html>
<head>
    <title>Palindrome Checker</title>
</head>
<body>

<h2>Check Palindrome String</h2>

<form method="post">
    Enter String:
    <input type="text" name="str">
    <input type="submit" value="Check">
</form>

<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $str = $_POST['str'];

    // Remove spaces and convert to lowercase
    $cleanStr = strtolower(str_replace(" ", "", $str));

    // Reverse string
    $revStr = strrev($cleanStr);

    echo "<h3>Result:</h3>";

    if ($cleanStr == $revStr) {
        echo "$str is a Palindrome String";
    } else {
        echo "$str is NOT a Palindrome String";
    }
}

?>
```

```
</body>
</html>
```

Q.1 Write CSS using HTML which displays following output Unordered lists

o Coffee

o Tea

o Milk

▪ Apple

▪ Mango

▪ Banana

Ordered list

I. Rose

II. Jasmine

III. Marigold

a. Sunflower

b. Tulip

c. Lily

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>List Styling Using CSS</title>
```

```
<style>
```

```
body {
    font-family: Arial;
}
```

```
.ul1 {
    list-style-type: circle; /* o style */
    color: brown;
}
```

```
.ul2 {
    list-style-type: square; /* ▪ style */
    color: green;
}
```

```
.ol1 {
    list-style-type: upper-roman; /* I, II, III */
    color: blue;
}
```

```
.ol2 {
    list-style-type: lower-alpha; /* a, b, c */
    color: purple;
}
```

```

    }
</style>

</head>

<body>

<h2>Unordered List</h2>

<ul class="ul1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

<ul class="ul2">
  <li>Apple</li>
  <li>Mango</li>
  <li>Banana</li>
</ul>

<h2>Ordered List</h2>

<ol class="ol1">
  <li>Rose</li>
  <li>Jasmine</li>
  <li>Marigold</li>
</ol>

<ol class="ol2">
  <li>Sunflower</li>
  <li>Tulip</li>
  <li>Lily</li>
</ol>

</body>
</html>

```

(15)

Q.2) Write a PHP script to print following floyd's triangle.

```

1
2 3
4 5 6
<?php

```

```

$n = 3; // number of rows
$num = 1;

```

```

for ($i = 1; $i <= $n; $i++) {

    for ($j = 1; $j <= $i; $j++) {
        echo $num . " ";
        $num++;
    }

    echo "<br>";
}

```

?>

Q.1) Write the JavaScript to convert temperature from Celsius to Fahrenheit.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Temperature Conversion</title>
```

```
</head>
```

```
<body>
```

```
<h2>Celsius to Fahrenheit Converter</h2>
```

Enter Celsius:

```
<input type="text" id="celsius">
```

```
<button onclick="convert()">Convert</button>
```

```
<p id="result"></p>
```

```
<script>
```

```
function convert() {
```

```
    let c = document.getElementById("celsius").value;
```

```
    // Convert to Fahrenheit
```

```
    let f = (c * 9/5) + 32;
```

```
    document.getElementById("result").innerHTML =
```

```
        "Fahrenheit: " + f;
```

```
}
```

```
</script>
```

```
</body>
</html>
```

Q.2) Write a PHP program to define Interface shape which has two method as area() and volume (). Define a constant PI. Create a class Cylinder implement this interface and calculate area and Volume.

```
<?php
```

```
// Define constant PI
define("PI", 3.14);
```

```
// Interface
interface Shape {
    public function area();
    public function volume();
}
```

```
// Cylinder class implementing interface
class Cylinder implements Shape {
```

```
    public $radius;
    public $height;
```

```
    function __construct($r, $h) {
        $this->radius = $r;
        $this->height = $h;
    }
```

```
    // Area =  $2\pi r(r + h)$ 
    public function area() {
        return 2 * PI * $this->radius * ($this->radius + $this->height);
    }
```

```
    // Volume =  $\pi r^2 h$ 
    public function volume() {
        return PI * $this->radius * $this->radius * $this->height;
    }
}
```

```
// Create object
```



```
$c = new Cylinder(5, 10);
```

```
echo "Surface Area of Cylinder: " . $c->area() . "<br>";
```

```
echo "Volume of Cylinder: " . $c->volume();
```

```
?>
```